SOFTWARE VALIDATION & VERSION HISTORY REPORT

PROJECT : *CalcSuite Pro: Digi-ToolKit*

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STATUS : *Final Release (v7.6)*

1. PROJECT OVERVIEW

CalcSuite Pro is a multi-tool mathematical workstation featuring secure Abstract Syntax Tree (AST) evaluation, Cramer's Rule matrices, and a custom polynomial dictionary engine for natural-language algebra parsing.

2. MASTER TEST CASES & VALIDATION SUITE

The following cases were executed to validate the engine's logical accuracy and error handling.

TAB 1: SCIENTIFIC CALCULATOR

- **Standard Arithmetic:** `2 * (5 + 5)`
  - **Expected Result:** `20`
- **Implicit Multiplier:** `2pi`
  - **Expected Result:** `6.2831853`
- **Trigonometry (DEG):** `sin(90)`
  - **Expected Result:** `1.0`
- **Advanced Logic:** `FIBS(10)`
  - **Expected Result:** `Popup window displaying 0, 1, 1, 2, 3, 5, 8, 13, 21, 34`
- **Syntax Guard:** `2x + 5`
  - **Expected Result:** `450ms human-speed delay -> Redirects to Algebra Mode.`

TAB 2: SOLVERS

- **Quadratic (Real):** `a=1, b=5, c=6`
  - **Expected Result:** `x1 = -2, x2 = -3`
- **Quadratic (Complex):** `a=1, b=2, c=5`
  - **Expected Result:** `x1 = -1 + 2i, x2 = -1 - 2i`
- **Linear Fallback:** `a=0, b=2, c=-10`
  - **Expected Result:** `Triggers 'Linear Fallback' note -> x = 5`
- **Linear 2x2 (Unique):** `EQ I: 2,3,10 | EQ II: 4,-1,6`
  - **Expected Result:** `x = 2, y = 2`
- **Linear (No Solution):** `EQ I: 2,4,10 | EQ II: 2,4,20`
  - **Expected Result:** `Triggers 'Parallel Lines / Inconsistent' system guard.`

TAB 3: SMART ALGEBRA

- **Numeric Bypass:** `5 + 10 * 2`
  - **Expected Result:** `Result: 25`
- **Simplification:** `2x + 5x - x`
  - **Expected Result:** `Simplified: 6x`
- **Linear (1 Var):** `3x - 5 = 10`
  - **Expected Result:** `Rearranged: 3x - 15 = 0 -> x = 5`
- **Linear (2 Var):** `2a - 6b = 19, 4a + 4b = 12`
  - **Expected Result:** `a = 4.625, b = -1.625`
- **Bracket Expansion:** `(x-2)(x+5)(x-1)(x+1) = 6`
  - **Expected Result:** `Polynomial iteration -> Rearranged -> +4 = 0`
- **Polynomial Guard:** `x^3 + 2x = 0`
  - **Expected Result:** `Triggers Degree-3 analytical limitation note.`
- **Cross-Term Guard:** `2xy + 5 = 0`
  - **Expected Result:** `Triggers multi-variable cross-term limitation note.`
- **Underdetermined System:** `7a+8b-9a+12b = 12a-13b+5a-8`
  - **Expected Result:** `Engine rearranges to -19a+33b+8 = 0 -> Triggers 'Underdetermined' guard -> Instructs user to provide a second equation.`
- **Format Guard (Non-Linear):** `(x-3)(x+4) = 2x`
  - **Expected Result: Triggers 'Format Error' ->** `Prompts user to move all variables to the left side to enforce standard polynomial form.`
- **Format Guard (Quadratic):** `x^2 + 4x = 2x + 7`
  - **Expected Result: Triggers 'Format Error' ->** `Prompts user to move all variables to the left side.`

TAB 4: HEALTH / BMI

- **Metric System:** `Age 25, Male, 100kg, 180cm`
  - **Expected Result:** `BMI: 30.9 (Obese) + Athlete context note.`
- **Imperial System:** `Age 25, Female, 140lb, 5ft 5in`
  - **Expected Result:** `BMI: 23.3 (Normal) + Female body-fat context note.`
- **Age Guard:** `Age 15, Male, 60kg, 160cm`
  - **Expected Result:** `Calculates BMI but overrides category -> requires chart.`

TAB 5: UNIT CONVERTER

- **Standard Conversion:** `Length -> 1 Metre to Centimetre`
  - **Expected Result:** `100 cm`
- **Advanced Conversion:** `Temperature -> 0 Celsius to Fahrenheit`
  - **Expected Result:** `32 °F`
- **Swap Functionality:** `Press 'Swap' on previous result`
  - **Expected Result:** `Swaps to 32 Fahrenheit -> 0 Celsius.`

3. EXTENDED VERSION HISTORY

This application underwent continuous daily iteration to refine security, logic handling, and user interface responsiveness.

Version	Date	Status	Key Upgrades & Architectural Changes
v1.0	30-03-2026	Legacy	Initial deployment. Basic arithmetic GUI logic established using standard libraries.
v1.2	01-04-2026	Legacy	Incremental logic optimizations and operational bug fixes.

v2.0	02-04-2026	Legacy	Major UI restructuring and visual layout adjustments.
v3.2	03-04-2026	Legacy	Multi-tool expansion. BMI and Unit Converter algorithms integrated.
v3.3	04-04-2026	Legacy	Refinement of converter parameters and unit scaling.
v4.0	07-04-2026	Legacy	Implementation of history tracking and state persistence.
v4.2	08-04-2026	Legacy	History UI optimizations and text formatting updates.
v5.0	09-04-2026	Legacy	Security & UI Overhaul. CustomTkinter implemented. Unsafe eval() replaced by strict AST parsing.
v6.0	10-04-2026	Legacy	Core solver engine mathematical foundation established.
v6.1	11-04-2026	Legacy	Quadratic solver edge-cases and imaginary number handling addressed.
v6.2	12-04-2026	Legacy	Linear 2x2 solver logic and Cramer's Rule matrix implementation.
v6.3	13-04-2026	Legacy	Error handling and validation deployment for the Solvers tab.
v7.0	15-04-2026	Stable	Smart Algebra logic introduced. Natural language parsing for single-variable equations.
v7.2	16-04-2026	Stable	Parsing improvements for algebraic terms and string extraction.

v7.3	20-04-2026	Stable	Simplification routing and numeric bypass logic integrated.
v7.4	21-04-2026	Stable	Iterative FOIL logic added for bracket expansion.
v7.5	22-04-2026	Stable	The "Pro" Overhaul. Digi-ToolKit branding applied. Smart cross-tab redirects built.
v7.6	24-04-2026	Final	**Stability Update:** Dictionary-based polynomial engine implemented. Epsilon floating-point tolerance deployed. Legacy code fully pruned. **UI Geometry Guard:** Patched character-overflow bugs in the Scientific Display (Consolas 28 font boundary) by optimizing the FIBS exception string to exactly 20 characters. **Symmetry Patch (Algebra):** Enforced syntax uniformity in the Dispatcher by mapping single-bracket exponential inputs (e.g., (x-3)^2) to a dedicated Quadratic Guardrail rather than falling back to generic Linear limits. **Parser Stability:** Hardened AST regex protocols to safely strip leading zeros (e.g., 0090) to prevent fatal Python octal-conversion crashes.